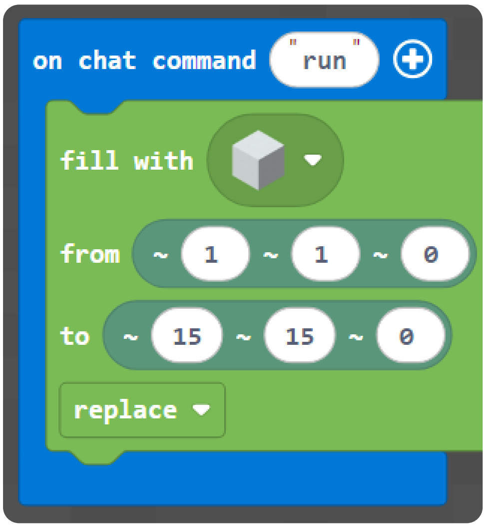
M003 Week 4 — English Worksheet

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Time:** about 45 minutes

This week you copy **two more pictures** onto your wall — a **cat face** and a **tree**. Same idea as last week: every block sits at a spot named by **two numbers — (x, y)**. x is how far **across**, y is how far **up**. Read the picture, find each block's (x, y), and place it.

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, 1, 0),
        pos(15, 15, 0),
        FillOperation.REPLACE)
player.on_chat("run", on_run)
```

> 🔴 **Big idea:** (x, y) is counted from your **home spot** — not from the corner of the world. Stand on the red block *every* time you run, or the whole picture shifts. Move your feet, move your picture.

1 · Predict

Read each set of steps. Before you imagine it happening, write what you think you will see.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
place brown block at (8, 4)
```

Four brown blocks. Do they make a line going across or going up? Which number stays the same? (This is the tree's trunk.)

```
place orange block at (2, 15)
place orange block at (3, 15)
```

Two black blocks near the top. Across or up? Which part of the cat could this be — an ear or a foot?

```
place green block at (8, 15)
place green block at (7, 14)
place green block at (9, 14)
```

These green blocks sit high and spread out as they go down. Do they make a fruit, or the top of the tree**?**

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — This should make the **tree trunk**: four brown blocks going **up** at x = 8, from (8, 1) to (8, 4). Right now they all land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Hint: to move up, the **y** has to change each time.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The tree trunk should grow **up** at x = 8: (8, 2), (8, 3), (8, 4). Right now it grows **across** instead.

```
place brown block at (2, 8)
place brown block at (3, 8)
place brown block at (4, 8)
```

Hint: check which slot the changing number is in — the x slot (across) or the y slot (up). The two numbers got swapped.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The cat should have **two eyes** at (6, 10) and (10, 10). Right now both eyes land on the **same side** of its face.

```
place green block at (5, 9)
place green block at (5, 9)
```

Hint: the second eye needs a different x.

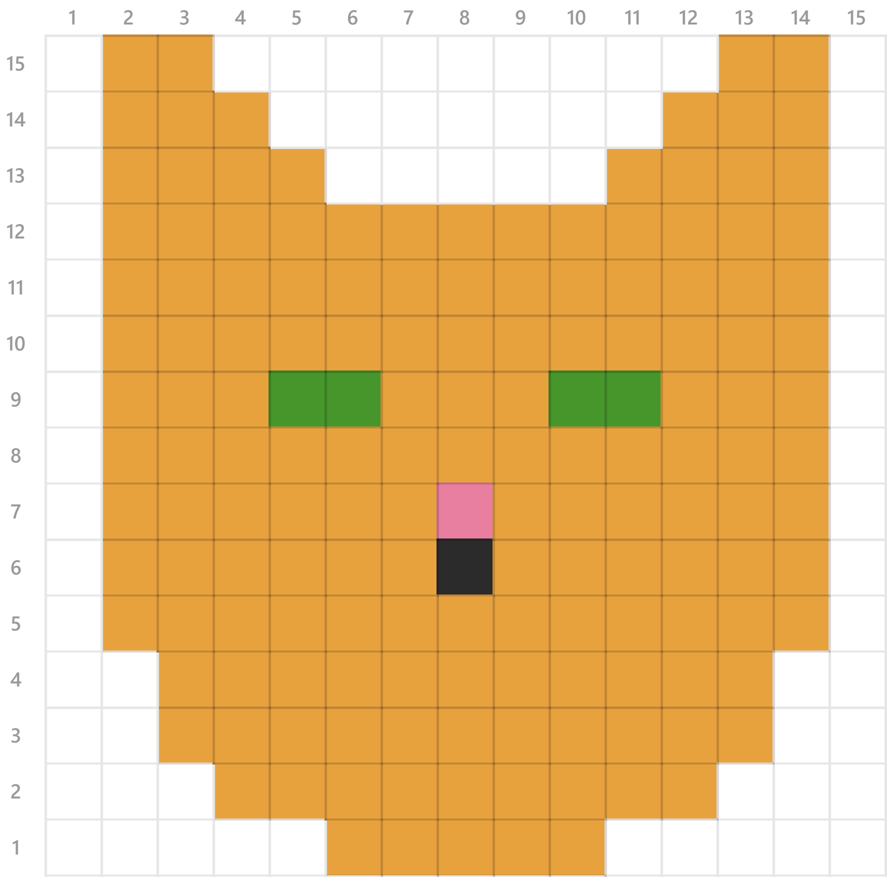
Write the fixed code:

Why was it wrong? Why does your fix work?

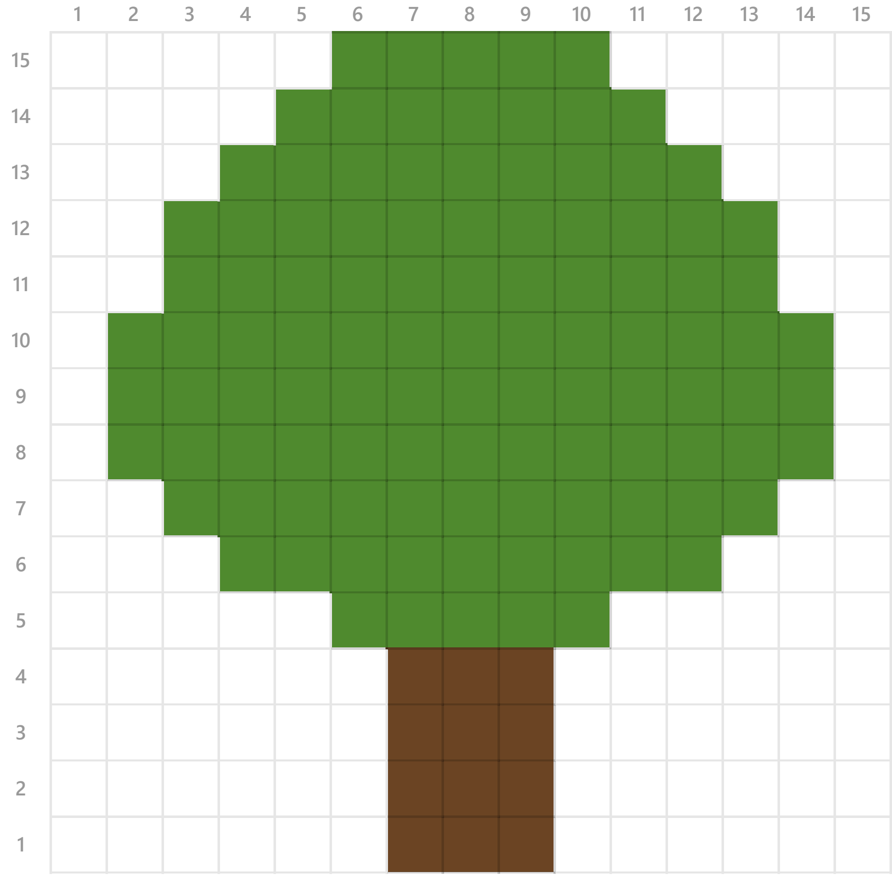
3 · Tell Me What You Built 📸

Now switch to your homework world and stand on your **home spot** (the red block). Pick **one** picture below and copy it onto your wall, block by block, by reading the (x, y) of every colored square. The numbers across the **top** are **x**, the numbers down the **side** are **y**.

🐱 Cat face



🌳 Tree



When your picture is finished, send a photo of it, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I chose to build the ...

The lowest block in my picture is at (x = ..., y = ...).

To go across a row I changed the ... number; to go up I changed the ... number.

The trickiest part to read off the grid was ...

One time my picture landed in the wrong place because ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you show your pixel art on the wall. Talk through it like you are teaching someone who has never seen it. Try to use these words: **x**, **y**, **coordinate**, **across**, **up**, **home spot**.

> 1. Show your home spot (the red block) and say why you stand on it every time.
> 2. Show your finished picture from the front.
> 3. Point at one block and read its coordinate out loud: "this one is at x ..., y ...".
> 4. Show one block that you placed in the wrong spot at first, and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✅

Send this worksheet + your walkthrough video to teacher on KakaoTalk.